Solvency Field Theory: Repaired Action Derivations Companion

Shaun Fosmark

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Status: mathematical support volume for the repaired live action candidate

Abstract

This companion preserves the derivation chain behind the repaired live action candidate for Solvency Field Theory. It records the honest source-sector variation, the minimal count-sector construction, the minisuperspace and non-FLRW gates, the exact static-vacuum obstruction of the absolute potential, the repair sequence that leads to the quadratic background-relative potential, the exact exterior branch-collapse result, and the repaired interior branch that yields an exact mass equation and a cleaned-up modified Tolman-Oppenheimer-Volkoff skeleton. Exact statements are separated from branch-level or effective-fluid closures.

1. Scope and live repaired action

The repaired effective action carried forward in the main document is

$$S^{\text{eff}} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \eta \, \nabla_\mu(\bar{\chi} + \psi) J^\mu_{\text{ops}} + \Lambda_C \mathcal{C}_{\text{min}} - V_{\text{quad}} + \mathcal{L}_m \right]. \tag{1}$$

The bookkeeping scalar is decomposed as

$$\chi(x) = \bar{\chi}(x) + \psi(x), \tag{2}$$

where $\bar{\chi}(x)$ is the cosmological/background clock branch and $\psi(x)$ is the local lag relative to that branch. The clock congruence is defined by the total clock field,

$$u^\mu = \frac{\nabla^\mu(\bar{\chi} + \psi)}{\sqrt{-\nabla_\alpha(\bar{\chi} + \psi)\nabla^\alpha(\bar{\chi} + \psi)}}, \qquad h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu. \tag{3}$$

The local operation density and current are

$$\rho_{\text{ops}} = \frac{T_{\mu\nu} u^\mu u^\nu + h^{\mu\nu} T_{\mu\nu}}{h}, \qquad J^\mu_{\text{ops}} = \rho_{\text{ops}} u^\mu. \tag{4}$$

The minimal count sector is

$$\mathcal{C}_{\text{min}} = \mathcal{H}_\chi^2 - \alpha(E_\chi + V_{\text{quad}}), \qquad \mathcal{H}_\chi = \frac{1}{3} \nabla_\mu u^\mu, \qquad E_\chi = T_{\mu\nu} u^\mu u^\nu, \tag{5}$$

with

$$\alpha = \frac{8\pi G}{3} \beta, \qquad \beta = \frac{1}{4\pi^2}. \tag{6}$$

The repaired potential is

$$\boxed{V_{\text{quad}} = V_0 \left(e^{-\psi} - 1\right)^2 .} \tag{7}$$

This companion explains why this is the live repaired action and what exactly it now does.

2. Honest source-sector variation

The decisive early action question was whether the hidden χ-dependence of the source term produced a pathological Euler-Lagrange equation. It does not. Define

$$X \equiv -\nabla_\mu \chi \nabla^\mu \chi > 0, \qquad s \equiv \sqrt{X}, \qquad u^\mu = \frac{\nabla^\mu \chi}{s}. \tag{8}$$

Holding the metric fixed during the χ-variation,

$$\delta X = -2\nabla_\mu \chi \, \nabla^\mu \delta\chi, \qquad \delta s = -u^\mu \nabla_\mu \delta\chi, \tag{9}$$

and the key identity is

$$\boxed{\delta u^\mu = \frac{1}{s} h^{\mu\nu} \nabla_\nu \delta\chi.} \tag{10}$$

Since $u_\mu \delta u^\mu = 0$, the variation of the clock congruence is purely spatial.

Introduce the clock-frame momentum density

$$q^\mu \equiv -h^\mu{}_\alpha T^{\alpha\beta} u_\beta, \qquad q^\mu u_\mu = 0. \tag{11}$$

Then the hidden variation of the source density collapses to

$$\boxed{\delta\rho_{\text{ops}} = -\frac{4}{hs} q^\mu \nabla_\mu \delta\chi.} \tag{12}$$

Now vary

$$\mathcal{L}_{\text{src}} = -\eta \, \nabla_\mu \chi \, J^\mu_{\text{ops}}. \tag{13}$$

Using $\delta J^\mu_{\text{ops}} = u^\mu \delta\rho_{\text{ops}} + \rho_{\text{ops}} \delta u^\mu$ one obtains, after integration by parts,

$$\boxed{\delta\mathcal{L}_{\text{src}} \mathrel{\hat{=}} \eta \left[\nabla_\mu J^\mu_{\text{ops}} + \frac{4}{h} \nabla_\mu q^\mu\right] \delta\chi.} \tag{14}$$

So the source contribution to the χ-equation is

$$\boxed{\eta \left[\nabla_\mu J^\mu_{\text{ops}} + \frac{4}{h} \nabla_\mu q^\mu\right].} \tag{15}$$

This is the first major simplification: the hidden variation is governed by clock-frame momentum flux, not by an uncontrolled tower of extra tensor terms.

3. Minimal count sector and first gates

The first viable local count candidate was

$$\boxed{\mathcal{C}_{\min} = \mathcal{H}_\chi^2 - \alpha(E_\chi + V).} \tag{16}$$

In homogeneous FLRW with

$$ds^2 = -N(t)^2 dt^2 + a(t)^2 d\mathbf{x}^2, \tag{17}$$

a homogeneous clock field gives

$$h^{00} = 0, \qquad \mathcal{H}_\chi = \frac{\dot{a}}{aN} = H, \tag{18}$$

so the count sector becomes

$$\mathcal{C}_{\min} = H^2 - \alpha(\rho + V). \tag{19}$$

That immediately yields the first gates:

- dust: if $V \to 0$, then $H^2 = \alpha\rho$;
- radiation: if $V \sim AH\rho$, then $H^2 = \alpha(\rho + AH\rho)$;
- vacuum: if $\rho = 0$ and $V \to 0$, then $H = 0$.

All three passed structurally. This was the first sign that the count sector could carry the $4\pi^2$ dust normalization without breaking the promoted radiation branch.

4. Variation of the count sector

The count-sector variation determines whether the count term injects hidden dynamics into sectors already under control. The variation of the clock expansion is

$$\delta\mathcal{H}_\chi = \frac{1}{3}\nabla_\mu \delta u^\mu = \frac{1}{3}\nabla_\mu \left(\frac{1}{s} h^{\mu\nu}\nabla_\nu \delta\chi\right), \tag{20}$$

so integrating the variation of $\mathcal{H}_\chi^2$ by parts twice gives

$$\left[\delta(\mathcal{H}_\chi^2)\right]_{\mathrm{EL}} = \frac{2}{3}\nabla_\nu \left(\frac{1}{s} h^{\mu\nu}\nabla_\mu \mathcal{H}_\chi\right)\delta\chi. \tag{21}$$

For the matter projection,

$$E_\chi = T_{\mu\nu} u^\mu u^\nu, \tag{22}$$

and, holding $T_{\mu\nu}$ fixed during the χ-variation,

$$\delta E_\chi = -\frac{2}{s} q^\mu \nabla_\mu \delta\chi. \tag{23}$$

So

$$[\delta(-\alpha E_\chi)]_{\mathrm{EL}} = -2\alpha\nabla_\mu \left(\frac{q^\mu}{s}\right)\delta\chi. \tag{24}$$

For a generic potential V, the total count-sector contribution is therefore

$$\boxed{\mathfrak{C}_\chi = \frac{2}{3}\nabla_\nu\left(\frac{1}{s}h^{\mu\nu}\nabla_\mu\mathcal{H}_\chi\right) - 2\alpha\nabla_\mu\left(\frac{q^\mu}{s}\right) + \alpha\frac{\partial V}{\partial\chi}.} \tag{25}$$

In homogeneous FLRW perfect-fluid sectors one has $q^\mu = 0$ and $h^{\mu\nu}\nabla_\mu\mathcal{H}_\chi = 0$, so the count variation collapses to the potential derivative alone. This is why the count term could remain active structurally without injecting hidden FLRW baggage.

5. Diffeomorphism identity and effective-fluid reading

The total action is diffeomorphism invariant. Writing the first variation schematically as

$$\delta S = \int d^4x\sqrt{-g}\left[\frac{1}{2}\mathcal{E}_{\mu\nu}\delta g^{\mu\nu} + \mathcal{E}_\chi\delta\chi + \mathcal{E}_A\delta\psi^A\right], \tag{26}$$

an infinitesimal diffeomorphism generated by ξ^μ yields the Noether identity

$$\boxed{\nabla^\mu\mathcal{E}_{\mu\nu} + \mathcal{E}_\chi\nabla_\nu\chi + \mathfrak{M}_\nu[\psi,\mathcal{E}_A] = 0.} \tag{27}$$

So on full shell the metric equation is automatically divergence-free. The action is structurally Bianchi-consistent; the practical question is whether one adopts a fully fundamental matter action or the effective-fluid reading. The derivations in this project use the effective-fluid reading, which matches the mesoscale bridge from discrete ticking to the fluid source density.

6. The $\varepsilon = 0$ branch and non-FLRW local operator

A major question was whether the bookkeeping kinetic regulator is actually needed. The minisuperspace audits showed that with $\varepsilon = 0$ the count constraint, lapse equation, and χ-equation remain algebraically sane. The surviving branch therefore behaves like a constrained bookkeeping-response branch rather than a propagating scalar-tensor branch.

Away from FLRW, take a local inertial patch with

$$\chi(t,x) = t + \varphi(t,x), \qquad U^\mu \approx (1, v^i), \tag{28}$$

and small velocity relative to the clock frame. To first order,

$$u^0 \to 1, \qquad u^x \to \partial_x\varphi, \qquad \mathcal{H}_\chi \to \frac{1}{3}\partial_x^2\varphi, \qquad q^x \to (\rho_0 + p_0)v. \tag{29}$$

The surviving local PDE takes the form

$$-\varepsilon\,\varphi_{tt} + \left(\varepsilon + \eta\frac{\rho_0 + 3p_0}{h}\right)\varphi_{xx} + \frac{2\Lambda_C}{9}\varphi_{xxxx} - (1+\Lambda_C\alpha)\bar{V}\,\varphi + \left(\frac{4\eta}{h} - 2\Lambda_C\alpha\right)\partial_x[(\rho_0+p_0)v] = 0. \tag{30}$$

This is the structural non-FLRW pass: the theory does not produce an uncontrolled transport sector, but a controlled local response equation with biharmonic stiffness and momentum-flux

forcing.

7. Static-vacuum obstruction of the absolute potential

The final obstruction to exact strong-field exterior closure turned out not to be the source sector or the count sector. It was the absolute potential. For a static spherically symmetric metric,

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{g(r)} + r^2 d\Omega^2, \tag{31}$$

consider a Killing-aligned clock branch

$$\chi = \chi_0 + \kappa t. \tag{32}$$

Then in vacuum,

$$\theta = 0, \qquad \mathcal{H}_\chi = 0, \qquad J^\mu_{\rm ops} = 0, \qquad E_\chi = 0, \tag{33}$$

so the source and count sectors are ready to die. But the absolute potential survives as

$$V_{\rm abs} = V_0 e^{-\chi_0 - \kappa t}, \tag{34}$$

which is explicitly time dependent. Consequently

$$\mathcal{C}^{\rm vac}_{\rm min} = -\alpha V_{\rm abs}, \tag{35}$$

and the reduced χ-equation still carries a term proportional to $V_{\rm abs}$. So the original action does not admit an exact static vacuum branch at finite χ.

8. Potential repair sequence

The repair sequence was forced by the hard gates:

$$V_{\rm abs} = V_0 e^{-\chi}, \tag{36}$$

$$V_{\rm rel} = V_0 e^{-(\chi - \bar{\chi})}, \tag{37}$$

$$V_{\rm shift} = V_0 \left(e^{-(\chi - \bar{\chi})} - 1 \right), \tag{38}$$

$$V_{\rm quad} = V_0 \left(e^{-(\chi - \bar{\chi})} - 1 \right)^2. \tag{39}$$

The first candidate fails because it is time dependent in static vacuum. The second removes the time dependence but leaves a nonzero vacuum load. The third makes the vacuum value vanish but still leaves a nonzero first derivative on the vacuum branch. The fourth is the first candidate satisfying both

$$V_{\rm quad}\big|_{\psi=0} = 0, \qquad \frac{dV_{\rm quad}}{d\psi}\Big|_{\psi=0} = 0, \tag{40}$$

while retaining positive restoring curvature,

$$\frac{d^2 V_{\rm quad}}{d\psi^2}\Big|_{\psi=0} = 2V_0 > 0. \tag{41}$$

So $V_{\rm quad}$ is the first repair candidate that solves both the metric-side and χ-equation-side vacuum obstruction.

9. Exact exterior branch collapse

On the repaired background-matched static vacuum branch,

$$\psi = 0, \qquad T_{\mu\nu} = 0, \qquad \theta = 0, \tag{42}$$

one has

$$V_{\rm quad} = 0, \qquad \frac{dV_{\rm quad}}{d\psi} = 0, \qquad J^{\mu}_{\rm ops} = 0, \qquad E_{\chi} = 0, \qquad \mathcal{C}_{\rm min} = 0. \tag{43}$$

So every non-Einstein sector vanishes simultaneously, and the repaired action collapses exactly to

$$\boxed{S_{\rm vac} = \int d^4x\sqrt{-g}\,\frac{R}{16\pi G}.} \tag{44}$$

Therefore the exterior vacuum equation is exactly

$$\boxed{G_{\mu\nu} = 0.} \tag{45}$$

For a static, asymptotically flat, spherically symmetric exterior, Schwarzschild follows exactly.

10. Minisuperspace survival after the repair

The repaired potential does not break the previously surviving branches.

For dust, on the background branch $\psi = 0$, the repaired potential vanishes, its first derivative vanishes, and the ψ-equation also vanishes. So the dust branch remains exact:

$$\boxed{H^2 = \alpha\rho.} \tag{46}$$

For radiation, the repaired potential is still zero on $\psi = 0$, but the radiation ψ-equation is nonzero, so radiation still drives the system away from the background branch. For vacuum, $\rho = 0$ and $\psi = 0$ give exact decoupling and $H^2 = 0$.

11. Static interior branch on $\psi = 0$

The repaired potential solves the exterior obstruction, but it does not force the matter interior to collapse automatically to GR. In TOV variables,

$$ds^2 = -e^{2\Phi(r)}dt^2 + \frac{dr^2}{A(r)} + r^2 d\Omega^2, \qquad A(r) = 1 - \frac{2m(r)}{r}, \tag{47}$$

with perfect fluid $T^\mu{}_\nu = \mathrm{diag}(-\rho, p, p, p)$, one finds on the repaired branch $\psi = 0$ that

$$u^t = e^{-\Phi}, \qquad u^r = 0, \qquad \theta = 0, \qquad \mathcal{H}_\chi = 0, \qquad V_{\mathrm{quad}} = 0, \tag{48}$$

while

$$E_\chi = \rho(r), \qquad \rho_{\mathrm{ops}} = \frac{\rho(r) + 3p(r)}{h}. \tag{49}$$

So the surviving reduced radial integrands are

$$I_m = -r^2 e^{\Phi} A^{-1/2} \rho, \tag{50}$$

$$I_{\mathrm{count}} = -\Lambda_C \alpha\, r^2 e^{\Phi} A^{-1/2} \rho, \tag{51}$$

$$I_{\mathrm{src}} = \frac{\eta\kappa}{h}\, r^2 A^{-1/2} (\rho + 3p). \tag{52}$$

The branch-level audit shows that the source sector does not enter the tt/lapse side directly, but does deform the radial side. The matter and count sectors combine into a simple rescaled matter load.

12. Exact mass equation and cleaned-up modified TOV skeleton

On the repaired $\psi = 0$ branch, the exact tt-side effective density is

$$\rho_{\mathrm{eff}}^{(tt)} = (1 + \Lambda_C \alpha)\rho, \tag{53}$$

so the exact mass equation is

$$\boxed{m'(r) = 4\pi r^2 (1 + \Lambda_C \alpha)\rho(r).} \tag{54}$$

The source-sector branch stress can be computed exactly from the reduced radial action. One finds

$$T^t{}_{t\,(\mathrm{src})} = 0, \qquad T^r{}_{r\,(\mathrm{src})} = -\frac{\eta\kappa}{h} e^{-\Phi(r)} \big(\rho(r) + 3p(r)\big). \tag{55}$$

Using perfect-fluid closure for the matter+count part gives the cleaned-up effective radial pressure

$$\boxed{p_{\mathrm{eff}}^{(r)} = (1 + \Lambda_C \alpha) p(r) - \frac{\eta\kappa}{h} e^{-\Phi(r)} \big(\rho(r) + 3p(r)\big).} \tag{56}$$

Therefore the repaired modified-TOV skeleton is

$$\boxed{\Phi'(r) = \frac{m(r) + 4\pi r^3 \left[(1+\Lambda_C\alpha)p(r) - \frac{\eta\kappa}{h} e^{-\Phi(r)}(\rho(r) + 3p(r))\right]}{r[r - 2m(r)]}.} \tag{57}$$

If separate matter conservation is retained in the effective-fluid reading,

$$\boxed{p'(r) = -(\rho(r) + p(r))\,\Phi'(r).} \tag{58}$$

This is the current live interior branch result.

13. Center regularity

A key question was whether the modified interior branch is actually regular at the stellar center. Expand

$$\rho(r) = \rho_c + \rho_2 r^2 + \cdots, \qquad p(r) = p_c + p_2 r^2 + \cdots, \tag{59}$$

$$\Phi(r) = \Phi_c + \Phi_2 r^2 + \cdots, \qquad m(r) = m_3 r^3 + m_5 r^5 + \cdots. \tag{60}$$

Substituting into the repaired modified-TOV skeleton produces no $1/r$ or constant singular terms. The first mass coefficients are

$$\boxed{m_3 = \frac{4\pi}{3}(1+\Lambda_C\alpha)\rho_c, \qquad m_5 = \frac{4\pi}{5}(1+\Lambda_C\alpha)\rho_2.} \tag{61}$$

The leading coefficients Φ_2 and p_2 solve cleanly, so the repaired interior branch is center-regular.

14. Weak-field / Newtonian limit

In the weak-field regime,

$$\frac{2m}{r} \ll 1, \qquad |\Phi| \ll 1, \qquad p \ll \rho, \qquad e^{-\Phi} \approx 1, \tag{62}$$

the repaired modified-TOV skeleton becomes

$$\boxed{m'(r) \approx 4\pi r^2 (1+\Lambda_C\alpha)\rho(r),} \tag{63}$$

$$\boxed{\Phi'(r) \approx \frac{m(r)}{r^2} - 4\pi\Gamma r\rho(r), \qquad \Gamma \equiv \frac{\eta\kappa}{h},} \tag{64}$$

$$\boxed{p'(r) \approx -\rho(r)\left[\frac{m(r)}{r^2} - 4\pi\Gamma r\rho(r)\right].} \tag{65}$$

So the count sector rescales the enclosed mass while the source sector adds a local correction to the Newtonian force law.

15. Status ledger

Exact.

- the repaired potential V_{quad},
- the exact static-vacuum obstruction of the absolute potential,
- the exact exterior vacuum branch collapse of the repaired action,
- exact Schwarzschild recovery in the exterior,
- the exact mass equation on the repaired $\psi = 0$ interior branch.

Branch-level / effective-fluid closed.

- source-sector structure and hidden-variation reduction,
- minimal count-sector structure,
- dust, radiation, and vacuum minisuperspace survival after the repair,
- non-FLRW local operator sanity,
- cleaned-up modified-TOV skeleton,
- center regularity,
- weak-field / Newtonian limit.

Still open.

- final covariant/background-independent packaging of the cosmological clock branch $\bar{\chi}$,
- full perfect-fluid variational closure of the radial sector,
- deeper interior phenomenology: matching, stability, and mass-radius predictions,
- perturbative normalization closure from the repaired action.

Closing remark

The repaired action is no longer a fragile Lagrangian ansatz. It is a live candidate action with an exact exterior GR branch and an explicit interior modified-TOV branch. The remaining work is now concentrated in fluid closure and phenomenology rather than in rescuing the basic action structure.